**A**

**Mini Project Report**

**on**

# SafeWave- An Emergency App

Submitted in partial fulfillment of the requirements for the

degree

**Second Year Engineering – Computer Science Engineering (Data Science)**

by


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# CERTIFICATE

This to certify that the Mini Project report on SafeWave App has been submitted by Vaishnavi Sawant (24107038), Sneha Shinde (24107069), Hastaksh Reddy (24107134) and Preshita Patil (24107126) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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# Chapter 1

## Introduction

In today's fast-paced world, emergency situations such as accidents, health crises, harassment, or personal safety threats can occur unexpectedly. During such critical moments, quickly informing trusted contacts and sharing location details can significantly improve the chances of receiving immediate help.

However, traditional methods of contacting people during emergencies—such as manually calling or messaging—can be slow and difficult, especially when the user is under stress or unable to operate their phone properly.

To address this issue, our mini project "SafeWave – Emergency SOS Mobile Application" is designed as a simple yet effective mobile safety tool. The application allows users to send an SOS alert to multiple trusted contacts instantly with their current location information by pressing a single button.

**1.1. Purpose:** The purpose of this document is to describe the design, development, and implementation of the SafeWave Emergency SOS Mobile Application.

This report provides detailed information about:

- The problem being addressed
- Objectives of the system
- Features and functionalities of the application
- System architecture and workflow
- Software requirements
- Project outcomes and results

The primary audience of this report includes project guides, evaluators, and students who want to understand the design and implementation of the SafeWave system.

**1.2. Problem Statement:** In emergency situations, individuals often struggle to quickly inform multiple trusted people about their situation. Manually calling or sending messages can take time and may not be possible if the person is in distress.

Some key challenges include:

- Difficulty in contacting several people at once during emergencies.
- Lack of quick communication systems that instantly alert trusted contacts.
- Delay in sharing the exact location of the person needing help.
- Increased risk due to slow response time.

**1.3. Objectives:** The main objectives of the SafeWave application are:

1. To develop a mobile application that allows users to send an SOS alert quickly.
2. To notify multiple trusted contacts simultaneously during emergencies.
3. To automatically fetch and share the user's current location in the alert message.
4. To provide a simple and user-friendly interface that can be easily used during stressful situations.
5. To enhance personal safety and emergency communication.

**1.4. Scope:** The SafeWave application focuses on improving emergency communication by providing a quick SOS alert system.

Scope of the project includes:

- Allowing users to store emergency contact details in the application.
- Providing a one-click SOS button to send emergency alerts.
- Fetching the current location using location services.
- Sending the location and emergency message to multiple contacts.
- Ensuring the application interface remains simple, fast, and accessible.

The system can be used by students, women, elderly people, and anyone who needs a quick safety alert system.

# Chapter 2

## Proposed System

The proposed system is a mobile application named SafeWave, designed to provide a quick emergency alert mechanism.

The application allows users to save trusted contacts in advance. In case of an emergency, the user can simply press the SOS button, which triggers the application to fetch the user's current location and send an emergency message to all saved contacts.

This system eliminates the need for manually calling or messaging people during stressful situations.

### 2.1 Features and Functionality

The SafeWave application provides the following features:

1. Emergency SOS Button: A large SOS button allows users to send emergency alerts instantly with a single tap.

2. Contact Management: Users can add, edit, or delete trusted emergency contacts within the application.

3. Location Sharing: The app automatically retrieves the user's current location using location services.

4. Address Generation: The location coordinates are converted into a readable address using geocoding techniques.

5. Multi-Contact Alert: The emergency message containing the user's location is sent to multiple saved contacts simultaneously.

6. User-Friendly Interface: The application provides a simple and clear interface that allows quick operation during emergencies.

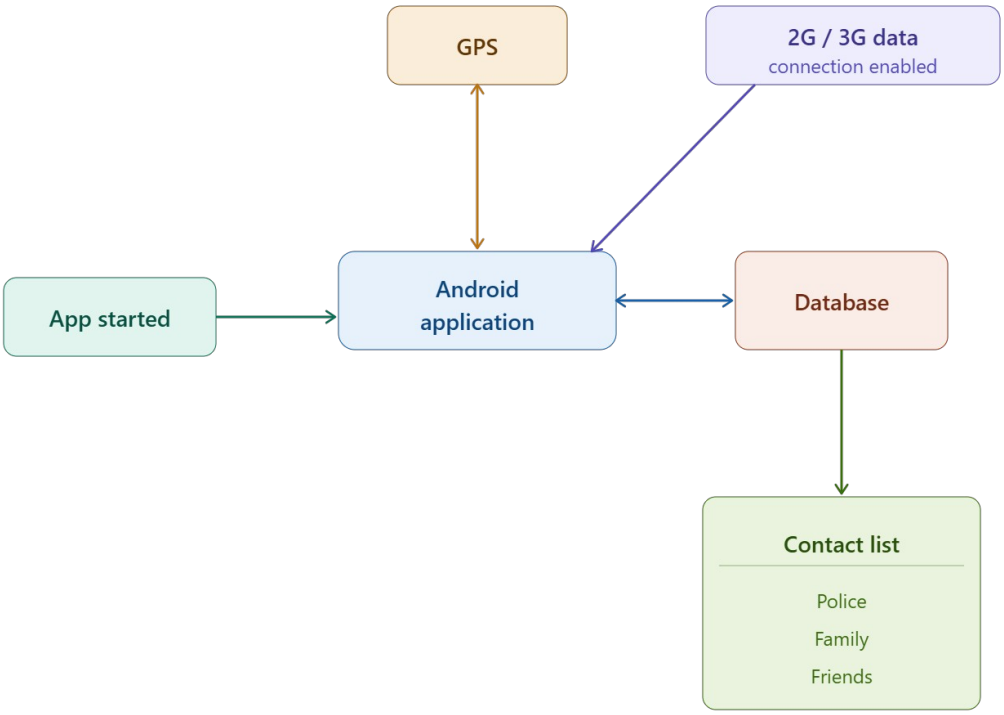


**Figure 1. Block representation of SafeWave App**

Figure 1- block diagram illustrates the working flow of the SafeWave SOS Application. It begins with the user entering contact details, including the name and phone number of trusted individuals. These details are then saved within the application, allowing the user to easily view and manage their contact list. The system provides options to add new contacts or display all saved contacts in a structured format. Users can also delete any contact if it is no longer needed. In case of an emergency, the user can press the “Send SOS” button. This action triggers the application to send an alert message to all saved contacts simultaneously. The overall design ensures a simple, quick, and efficient process for emergency communication.

# Chapter 3

## Project Outcomes

The SafeWave project successfully achieved the following outcomes:

- Development of a functional SOS emergency mobile application.
- Successful storage and management of multiple emergency contacts.
- Implementation of location detection and address generation.
- Sending SOS alerts to multiple contacts simultaneously.
- Creation of a simple and intuitive user interface.

The application demonstrates how mobile technology can be used to enhance personal safety and emergency communication.

**1. Development of a Functional SOS Emergency Mobile Application**

The SafeWave project resulted in the successful development of a fully functional Android-based mobile application designed specifically for emergency situations. The application integrates multiple features such as contact management, location tracking, and SOS alert triggering into a single platform. It ensures that users can access emergency functionalities quickly without navigating through complex menus. The app is designed to work efficiently even under stress, making it reliable during critical situations.

**2. Successful Storage and Management of Multiple Emergency Contacts**

The application provides a dedicated feature for storing and managing emergency contacts. Users can add, view, and delete multiple contacts as per their requirements. These contacts are securely stored within the app's local database, ensuring quick retrieval when needed. The management system is designed to be simple and user-friendly, allowing users to update their contact list easily without any technical knowledge.

### 3. Implementation of Location Detection and Address Generation

One of the key features implemented in SafeWave is real-time location detection. The application uses device location services (such as GPS) to fetch the user's current coordinates accurately. These coordinates are then converted into a readable address using geocoding techniques. This ensures that the emergency message contains precise and understandable location details, making it easier for contacts to locate the user quickly.

### 4. Sending SOS Alerts to Multiple Contacts Simultaneously

The core functionality of the SafeWave application is its ability to send SOS alerts instantly to multiple contacts at the same time. With a single tap on the SOS button, the app automatically composes and sends an emergency message along with the user's location details. This eliminates the need to contact each person individually, significantly reducing response time and improving the chances of receiving timely help.

### 5. Creation of a Simple and Intuitive User Interface

The application is designed with a focus on simplicity and ease of use. The interface includes clear buttons, minimal steps, and an organized layout to ensure quick navigation. Special attention is given to the SOS button, making it easily accessible and noticeable. This intuitive design ensures that users can operate the app efficiently even in stressful or panic situations, which is crucial for an emergency application.

# Chapter 4

## Software Requirements

The SafeWave application requires the following software tools and technologies:

| Software | Purpose |
|---|---|
| Android Studio | Mobile application development |
| Java | Programming language for application logic |
| Google Location Services API | Fetching user's current location |
| Android SDK | Development framework for Android apps |
| Emulator /Android Device | Testing the application |

**Table 4.1: Software Requirements for SafeWave**

Table 4.1 shows a set of essential software tools and technologies to ensure efficient development, smooth execution, and reliable performance. The primary development environment used for this project is Android Studio, which provides a comprehensive platform for designing, coding, testing, and debugging the application. It offers an integrated interface that supports XML-based user interface design and Java-based programming, making it suitable for building Android applications. The core programming language used in the development of SafeWave is Java, which is responsible for implementing the application logic, handling user inputs, managing contact data, and executing the SOS alert functionality. Java’s object-oriented nature allows for modular and structured coding, which improves maintainability and scalability of the application.

# Chapter 5

## Project Design

The system design consists of the following main components:

1. User Interface Module – Allows users to interact with the application.
2. Contact Management Module – Stores emergency contacts.
3. Location Detection Module – Fetches the user's GPS location.
4. Geocoding Module – Converts location coordinates into an address.
5. SOS Alert Module – Sends emergency messages to all saved contacts.

Workflow of the System

1. User enters contact details.
2. Contacts are saved in the application.
3. User presses the SOS button.
4. Application fetches the current location.
5. Location is converted into a readable address.
6. SOS message with location is sent to all contacts.

**1. User Interface Module**

The User Interface (UI) Module is responsible for enabling interaction between the user and the SafeWave application. It provides a clean, simple, and intuitive layout that allows users to easily navigate through the app's features. The UI includes input fields for entering contact details, buttons for saving and viewing contacts, and a prominently placed SOS button for emergencies. Special attention is given to accessibility and ease of use so that even in panic situations, users can quickly perform actions without confusion. The design minimizes steps and ensures all important functions are easily reachable.

**2. Contact Management Module**

The Contact Management Module handles the storage and organization of emergency contacts. Users can add new contacts by entering names and phone numbers, view the list

of saved contacts, and delete contacts when required. These contacts are stored locally within the application using a database or storage system, ensuring quick access during emergencies. This module ensures that users do not have to manually enter contact details every time an SOS alert is triggered, thereby saving valuable time.

**3. Location Detection Module**

The Location Detection Module is responsible for obtaining the user's real-time location. It uses device location services such as GPS or network-based location providers to fetch accurate latitude and longitude coordinates. This module works in the background when the SOS button is pressed, ensuring that the location is captured instantly. Accurate location detection is crucial for helping responders or contacts reach the user quickly during emergencies.

**4. Geocoding Module**

The Geocoding Module converts the raw latitude and longitude coordinates obtained from the location detection module into a human-readable address. This is done using geocoding techniques or APIs (such as Google Location Services). Instead of sending complex coordinates, the app provides a proper address (e.g., street name, area, city), which makes it easier for contacts to understand the user's location. This module improves clarity and usability of the SOS message.

**5. SOS Alert Module**

The SOS Alert Module is the core functionality of the SafeWave application. When the user presses the SOS button, this module is activated. It collects the user's current location (from the location module), converts it into an address (via the geocoding module), and composes an emergency message. This message is then sent simultaneously to all saved emergency contacts using messaging services. The module ensures fast and reliable communication, which is critical in emergency situations.

# Chapter 6

## Project Scheduling

The project was scheduled over a period of four months from July to October.

| Month | Task |
|---|---|
| July | Project idea selection and requirement analysis |
| August | Application design and UI development |
| September | Implementation of SOS and location features |
| October | Testing, debugging, and documentation |

**Table 6.1: Project Schedule**

Table 6.1 shows the Project schedule wherein the development of the SafeWave Emergency SOS Mobile Application was systematically planned and executed over a structured timeline to ensure smooth progress and timely completion. The project schedule was divided into two major phases: Project Creation and Initiation and Project Design and Implementation. Each phase consisted of multiple tasks that were carefully organized across several weeks.

The first phase, Project Creation and Initiation, focused on laying the foundation for the project. It began with group formation and the selection of the project topic. During this stage, the idea of developing an emergency SOS application was finalized after analyzing real-world safety concerns and identifying the need for quick communication during emergencies. Following this, a detailed problem statement was defined, highlighting the limitations of existing emergency communication methods and the necessity for a faster, more efficient solution.

Once the problem was clearly identified, the next step involved discussing the project scope and gathering requirements. This included identifying key features such as contact management, SOS alert functionality, and location tracking. A basic system design was then created to visualize how the application would function. This design phase included outlining the workflow, defining modules, and preparing initial diagrams such as flowcharts and architecture diagrams. The first phase concluded with an initial presentation, where the proposed idea and design were reviewed.

The second phase, Project Design and Implementation, focused on transforming the conceptual design into a working application. This phase began with database design, where the structure for storing emergency contacts was created. The database ensured efficient storage, retrieval, and management of contact information within the app.

Following this, the application development process began. The user interface was designed to be simple, intuitive, and easy to use, especially during emergency situations. Features such as adding contacts, viewing saved contacts, and deleting contacts were implemented. Special attention was given to the SOS button, ensuring it was prominently placed and easily accessible.

Next, the core functionality of the application was developed, including integration of all modules. The location detection feature was implemented using device location services to fetch real-time coordinates. These coordinates were then converted into a readable address using geocoding techniques. The SOS alert module was integrated to send emergency messages along with location details to all saved contacts simultaneously.

After the development phase, the project underwent testing and debugging. Various test cases were executed to ensure that all features worked correctly and efficiently. Issues related to location accuracy, message delivery, and user interface responsiveness were identified and resolved during this stage.

Finally, the project concluded with a second presentation and documentation phase. The complete working model of the SafeWave application was demonstrated, highlighting its features and functionalities. Proper documentation was prepared, including all design details, implementationsteps, and results.

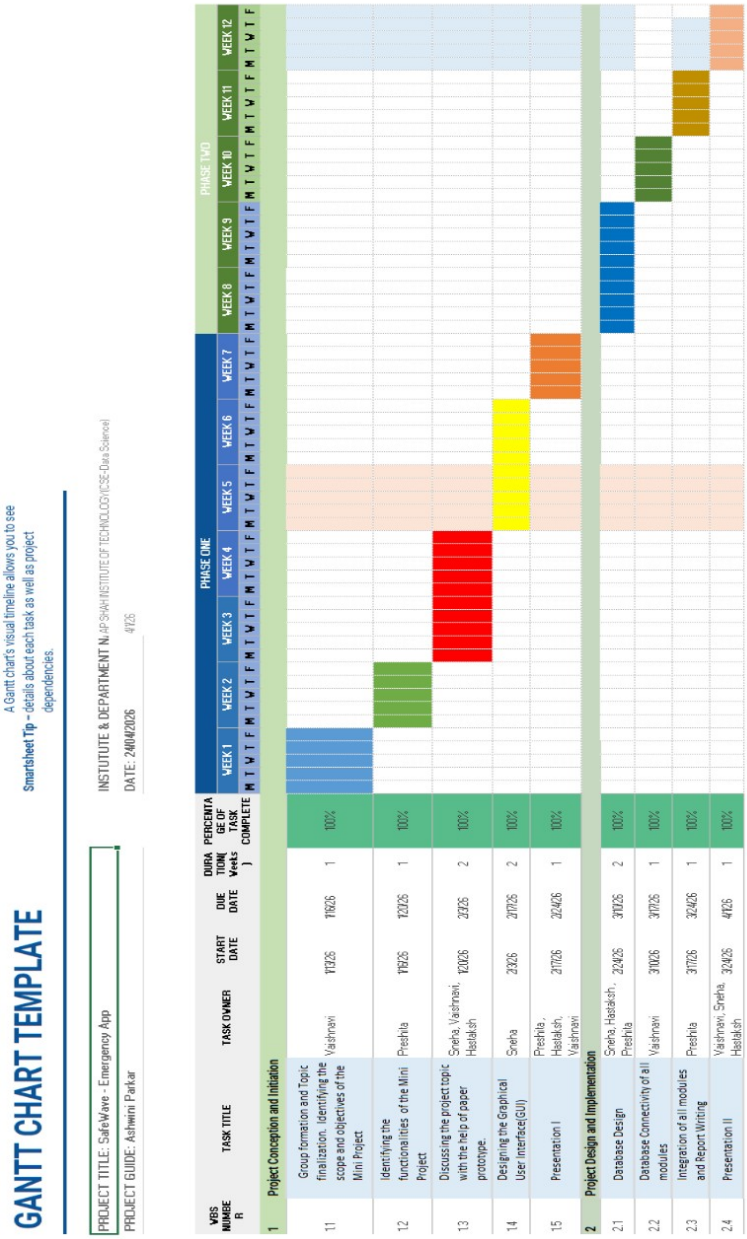


**Figure 2. Gantt Chart of SafeWave**

# Chapter 7

## Results

The SafeWave application was successfully implemented and tested.

The results obtained include:

- Emergency messages were successfully sent to multiple contacts.
- User's current location was accurately fetched using location services.
- The generated address was correctly included in the SOS message.
- The application interface was easy to navigate and responsive.

Testing confirmed that the system works effectively as a quick emergency communication tool.

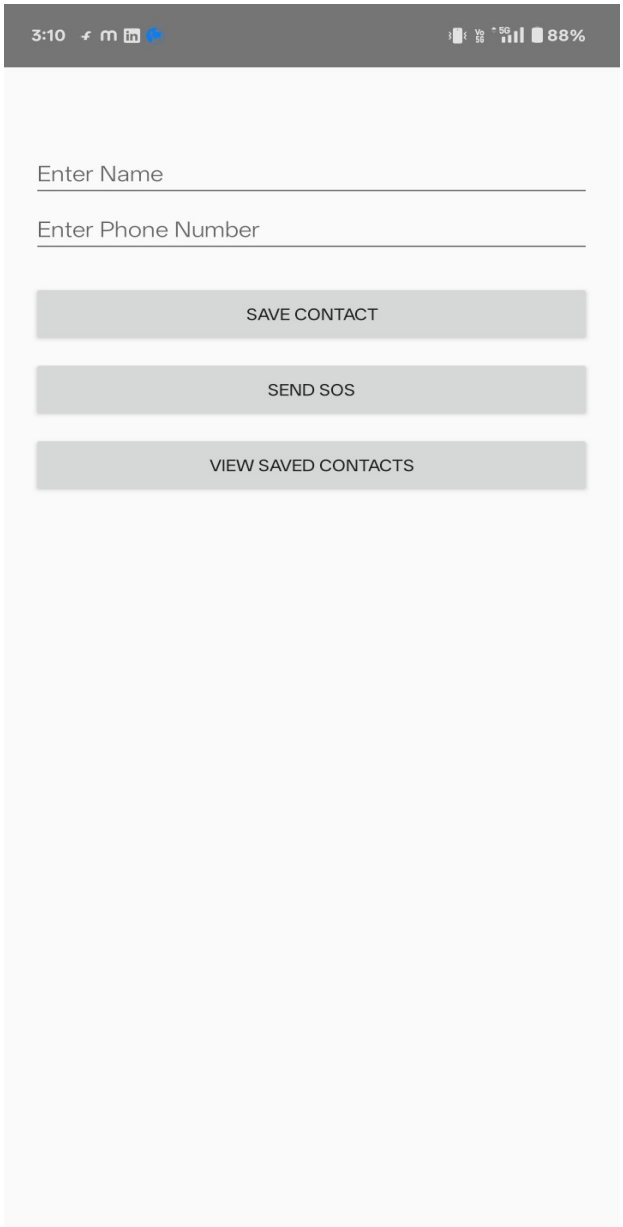


**Figure 3. Main Interface Screen of SafeWave**

Figure 3 represents the- Contact Input and Main Interface Screen

This image represents the main interface of the SafeWave SOS application, where users can input and manage their emergency contacts. The screen is designed with simplicity and clarity, featuring two primary input fields labeled “Enter Name” and “Enter Phone Number.” These fields allow users to easily add contact details without confusion. Below the input section, three main buttons are provided: “Save Contact,” “Send SOS,” and “View Saved Contacts.” Each button is clearly labeled and spaced properly to ensure easy accessibility, especially during emergency situations.

The “Save Contact” button allows users to store entered details in the application, while the “Send SOS” button is the key feature used to trigger emergency alerts. The “View Saved Contacts” button enables users to access and review the list of previously saved contacts. The minimalistic design reduces complexity and ensures that users can operate the app quickly, even under stress. Overall, this screen demonstrates a user-friendly layout that prioritizes speed, accessibility, and ease of use, which are essential for an emergency application.

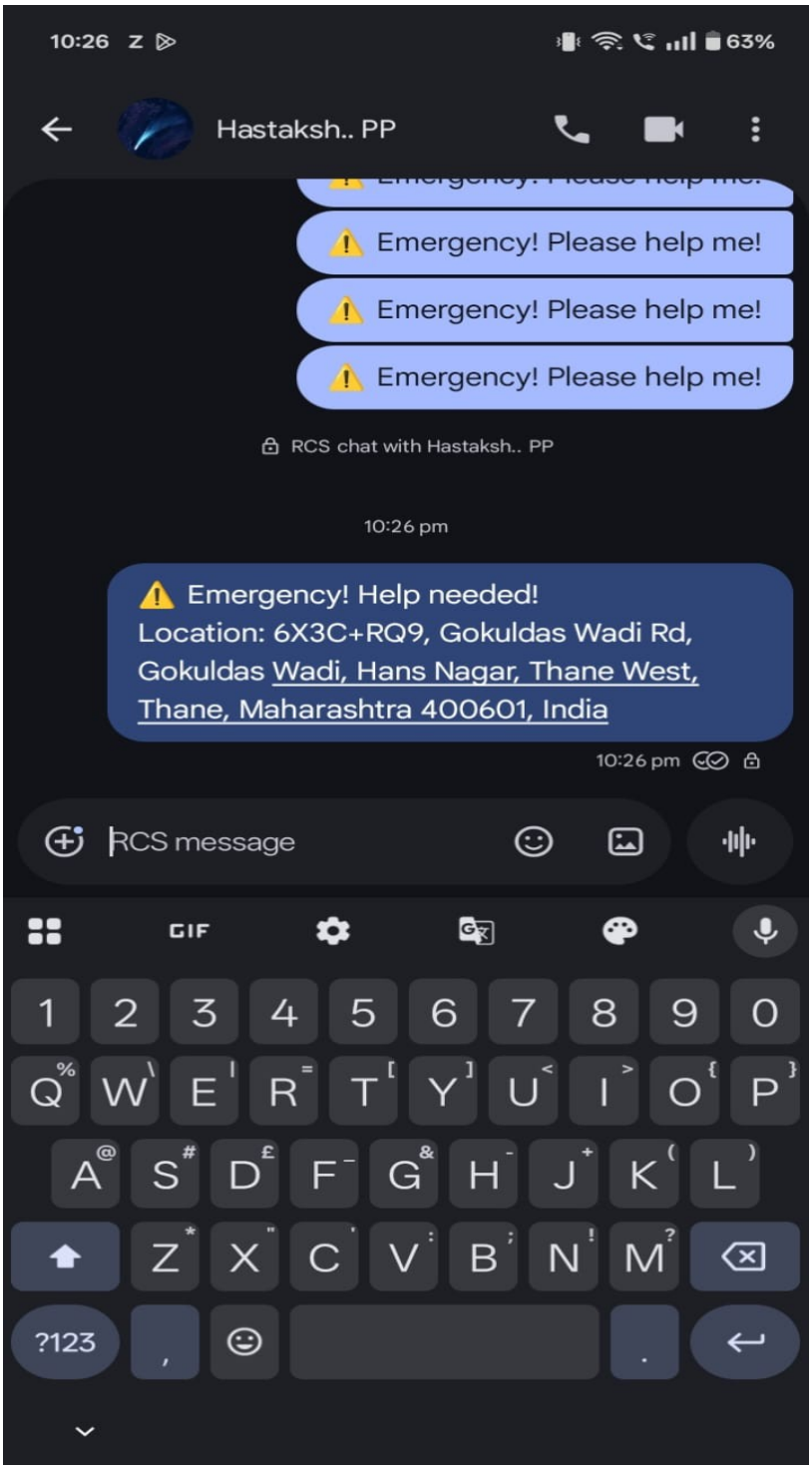


**Figure 4. Location sharing in SafeWave**

Figure 4 represents the- SOS Message Sent with Location

This image displays the result of the SOS functionality in the SafeWave application, showing how emergency messages are delivered to contacts. The screenshot is taken from a messaging interface where multiple alert messages have been sent. The messages include a warning symbol and text indicating an emergency situation, such as “Emergency! Help needed!” to immediately capture attention.

A key highlight of this screen is the inclusion of the user’s current location in the message. The location is presented in a readable address format, making it easy for the recipient to understand where the user is located. This feature is crucial in emergency situations, as it allows contacts to respond quickly or share the information with authorities if needed.

The repeated messages ensure that the alert is noticed, increasing the chances of a quick response. This image demonstrates the effectiveness of the SafeWave application in delivering timely and informative alerts. It confirms that the SOS feature works successfully by combining urgent messaging with accurate location sharing.

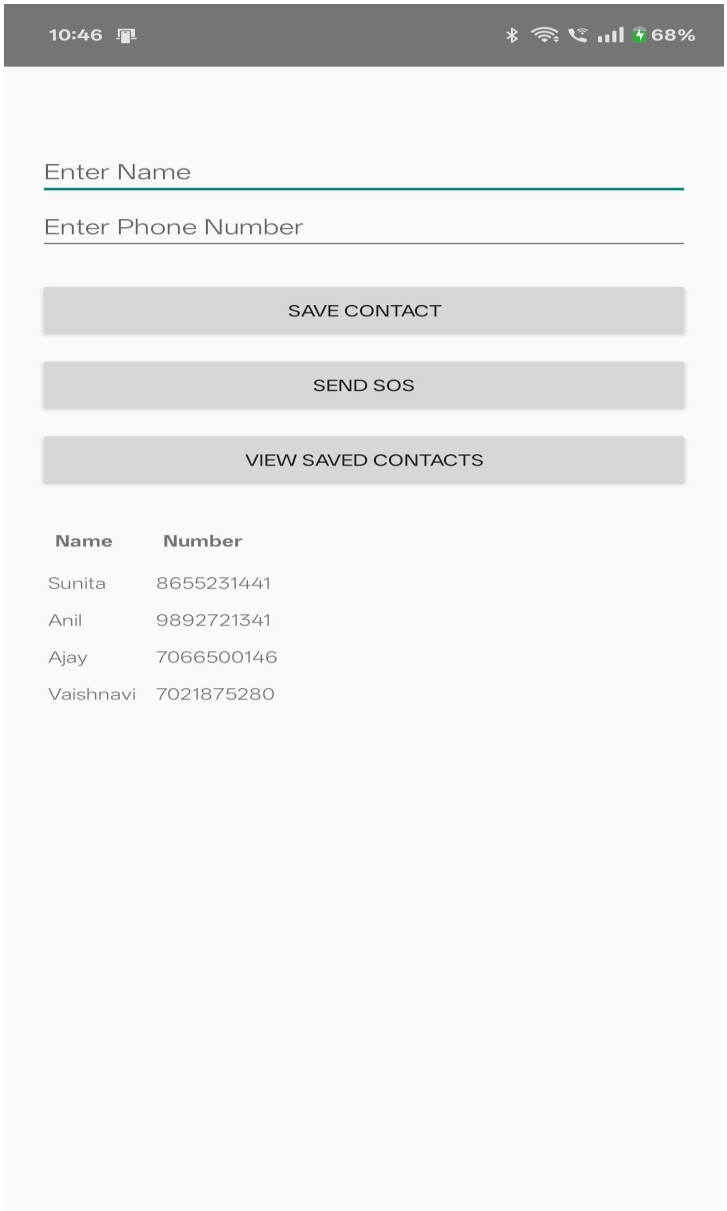


**Figure 5. Saved Contacts in SafeWave**

Figure 5 represents the- View Saved Contacts Screen

This image shows the “View Saved Contacts” functionality of the SafeWave application. After saving multiple emergency contacts, users can access this screen to view all stored entries in a structured format. The contact list is displayed in a simple tabular form with two columns: “Name” and “Number.” This clear presentation allows users to quickly identify and verify saved contacts.

The screen also retains the main interface elements such as the input fields and action buttons, ensuring consistency in design. Users can easily add new contacts or send an SOS alert directly from this screen. The listed contacts demonstrate that the application successfully stores multiple entries and retrieves them when required.

This feature is important for managing emergency contacts efficiently, as users may need to update or review their list periodically. The clean layout and readable text enhance usability and ensure that users can navigate the contact list without difficulty. Overall, this screen highlights the effectiveness of the contact management system within the SafeWave application.

# Chapter 8

## Conclusion

The SafeWave Emergency SOS Mobile Application provides a reliable solution for emergency communication. By allowing users to instantly alert trusted contacts and share their location with a single tap, the application improves response time during critical situations.

The project demonstrates the practical use of mobile technologies to enhance personal safety. SafeWave can be particularly helpful for students, women, elderly individuals, and people traveling alone.

With further improvements such as live tracking, voice-activated alerts, and integration with emergency services, the application can become an even more powerful safety tool.

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